

1. Key Challenges: Threats determined through SWOT Analysis



PESTEL Analysis

- Used to analyse external environment for Threats (here named 'Threat')

1. **Political** Threats:

- **Regulatory Compliance:** must adhere to varying global regulations. E.g. California Consumer Privacy Act (CCPA) & GDPR.
- **Trade Policies and tariffs:** can impact sourcing of components and resources. E.g. European Union's Digital Markets Act (DMA).

2. **Economic** Threats:

- **High Development Costs:** significant investment in R&D needed.
- **Highly competitive market**

3. **Social** Threats:

- **Consumer Acceptance:** Customers accustomed to existing systems (e.g., Apple CarPlay, Android Auto) may resist.

4. **Technological** Threats:

- **Rapid Technological Advances:** can quickly render new systems obsolete.
- **Cybersecurity Risks:** pose risks to user and data security.

5. **Environmental** Threats:

- **Sustainability Concerns:** pressure to develop environment-friendly technologies.

6. **Legal** Threats:

- **Intellectual Property Issues:** potential legal disputes over IPR.
- **Compliance with Data Protection Laws:** E.g. GDPR.

McKinsey's 7Ss Framework

- Used to analyse internal environment for Threats (here named 'Weakness')

1. **Strategy**

- **Misalignment with Technological Trends:** like 5G or AI
- **Long Development Timelines:** lead to missed opportunity to capture the market and new customer segments.

2. **Structure**

- **Siloed Developments**
- **Integration Challenges:** Difficulty merging traditional automotive structures with software-focused development processes required for MB.OS.

3. **Systems**

- **Inadequate IT Infrastructure:** Legacy IT infrastructure might hinder the full-scale deployment and operation of MB.OS.

4. **Shared Values**

- **Cultural Resistance to Change:** resistance to new ways of working & ideas.

5. **Style**

- **Lack of Agile Methodologies:** may hinder efficient and flexible development of MB.OS.

6. **Staff**

- **Skill Gaps in Software Development:** Lack of skilled software developers.

7. **Skills**

- **Insufficient Expertise in AI & Machine Learning:** could slow down the development and optimization of advanced features in MB.OS
- **Outdated Engineering Skills**

- The External Threat analysis (PESTEL) and the Internal Threat analysis (McKinsey 7S) together provide a holistic view of the challenges Mercedes faces in developing its own Operating System MB.OS.

2. MB.OS Customer Benefit – Navigation Service

Aim: To calculate the potential revenue in Euros for the year 2025 generated by a new and advanced navigation service, available for cars utilizing the new MB.OS operating system.

Potential Revenue = Price * Quantity Sold

Status Quo & Assumptions:

- As of today, the MB.OS will be supported only on the new electric first platform – [Mercedes Modular Architecture \(MMA\)](#).
 - Therefore, [only Battery Electric Vehicles \(BEVs\)](#) based on [MMA](#) are considered for calculations (4 new all-electric vehicles debuting in 2025).
- **BEVs Sales (2025)**: Same trend as in 2024 assumed due to slow adoption of BEVs; thus, BEVs sales in 2025 – **185,100 BEVs**.
- **MMA share of BEVs**: Assumed **15%**; due to gradual rollout, limited availability (4 initial all-electric) & slow early adoption.
- **Adoption Rate for Navigation Service**: Assumed **16%** adoption rate (due to hesitance to adopt new technologies) [including innovators & early adopters](#); based on the Diffusion of Innovations Theory. [Early Majority Theory: Stages, Examples and Types](#)
- Current/Recent Pricing of Navigation Package: €15/month, €139/12 months, €329/36 months.
 - **New service pricing** based on current/recent pricing: **€20/month, €150/12 months, €350/36 months** (higher price reflects more advanced technology, enhanced features and value proposition of the new service).
 - Revenue Calculation Method: Monthly revenue multiplied by 12; 36-month revenue divided by 3 to normalize for annual comparison.

2. MB.OS Customer Benefit – Navigation Service

Revenue
(in Euros in 2025 from sale of new
Navigation Service for MMA using
BEVs)

=

Price
(per Navigation Package
in Euros)

*

Quantity sold
(of MMA using BEVs with
new Navigation Service)



$185,100 * 0.15 * 0.16$
= 4442 cars

(rounded down as fractional cars not possible)

① **62,180 €**
(revenue per month from MMA using BEVs from new
Navigation Service)

=

20 €
(subscription price per month)

*

$0.70 * 4442 = 3109$ cars using monthly subscription
(rounded down as fractional cars not possible)

② **133,200 €**
(revenue per 12 months from MMA using BEVs from
new Navigation Service)

=

150 €
(subscription price per 12 months)

*

$0.20 * 4442 = 888$ cars using 12 month subscription
(rounded down as fractional cars not possible)

③ **155,400 €**
(revenue per 36 months from MMA using BEVs from
new Navigation Service)

=

350 €
(subscription price per 36 months)

*

$0.10 * 4442 = 444$ cars using 36 month subscription
(rounded down as fractional cars not possible)

Normalizing above
Revenues to reflect 12
month period

①
 $62,180 € * 12 = 746,160 €$
(revenue per 12 months from MMA using BEVs using
monthly subscriptions from new Navigation Service)

,

②
133,200 € (no need for normalization)
(revenue per 12 months from MMA using BEVs using
12 month subscriptions from new Navigation Service)

,

③
 $155,400 / 3 = 51,800 €$
(revenue per 12 months from MMA using BEVs using
36 month subscriptions from new Navigation Service)

Revenue

=

746,160 €

+

133,200 €

+

51,800 €

=

931,160 € is the potential revenue in 2025 from introducing an advanced navigation service

3. Business Case Optimization Navigation Service

- An unfavorable business case could imply a poor profit performance, which can be improved by addressing either **revenue** or **cost** factors, or a combination of both (since $\text{Profit} = \text{Revenue} - \text{Costs}$).
- In addition, other **qualitative factors** impacting the service's attractiveness and impact must also be considered to achieve a more favorable business case.
- Following are levers to consider to optimize the Business Case:

Revenue	Costs	Qualitative Factors
• Increase price • Price Premium for Advanced Features – Charging more for advanced capabilities. • Dynamic Pricing Strategies – based on demand & competition. • Tiered Pricing – multiple subscription tiers with differentiated features targeting a wider range of customers • Increase quantity sold • Targeted Marketing for Early Adopters & Innovators • Bundling with Other Services/Packages • Expansion into New Markets – geographic or other segments	• Reduce fixed costs • R&D Costs – Prioritize spending on high-impact developments; leverage existing technologies • Infrastructure Costs – Optimize production and server infrastructure; transition to cost-effective cloud hosting • Administrative Overheads – Streamline operations to lower overhead. • Reduce variable costs • Production Costs – Streamlining development, reducing material waste, and optimizing resources.	• Buying Criteria: Focus on ease of use, safety, and personalization to boost adoption and customer loyalty. • Competition: Monitor competitors' advancements closely to ensure we can address gaps in their offerings and maintain a competitive edge. • USP: : Differentiate with exclusive MB.OS features to justify premium pricing and make it stand out in the market. • Market Trends: Leverage sustainability and tech trends (e.g., eco-friendly routes, AI integration) to increase adoption and revenue.